



Assessing DIII Basketball Lineup Effectiveness

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BACKGROUND

- CMU's Division III Women's Basketball team primarily used either a 3-Guard (3G) or a 4-Guard (4G) lineup during its 2024-2025 season
- We scraped possession-based data from the box score and play-by-play recaps of games from CMU's Women's Basketball's 2024-2025 season
- Primary Goal:
 - Evaluate whether 3G and 4G lineups differed in offensive efficiency, shooting, rebounding, and overall effectiveness
- This will provide Coach McConnell-Miller with insights as to when a post-heavy (3G) or a guard-heavy (4G) lineup would be best to use

DATA

Extracting data:

- Python code was written to scrape play-by-play game data (points, possessions, score differential, quarter, time, number of guards, fga, fta, 3pa) from the CMU Athletics website
- Each observation corresponds to a unique lineup segments
- 27 games in 2024-2025 season, 198 unique lineups (61 of 3G, 123 of 4G)

Processed data:

- Non-competitive minutes were filtered out based on thresholds derived from the top and bottom 10% of the score differential and total seconds ECDF
- Merged lineups at the individual game and season level and calculated metrics
- After filtering lineups, left very little data: 5 unique 3G lineups and 14 unique 4G lineups

Predictor Variables: Lineup Type

Outcome Variables: Performance Metrics

Net Rating	Offensive Rating - Defensive Rating	True Shooting % (TS%)	$100 * (PTS / (2 * (FGA + 0.44 * FTA)))$
Offensive Rating	$100 * (pts / possessions)$	True Rebounding % (TRB%)	$100 * (Team Rebounds / Total Rebounds)$
Defensive Rating	$100 * (Opponent pts / opponent possessions)$	3PA/FGA	3PA / FGA
Pace	$40 * ((Possessions + Opponent Possessions) / (2 * Lineup Seconds / 60))$		

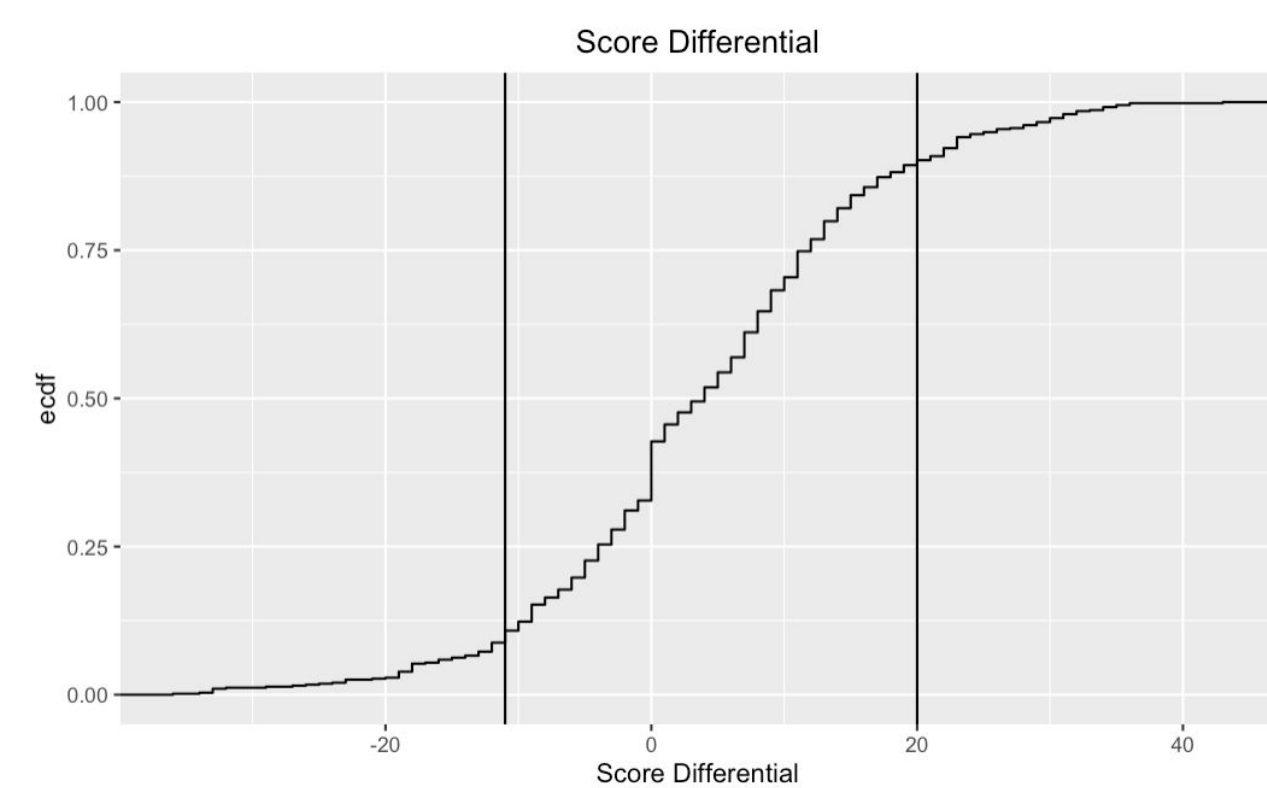


Figure 1: ECDF of Score Differential

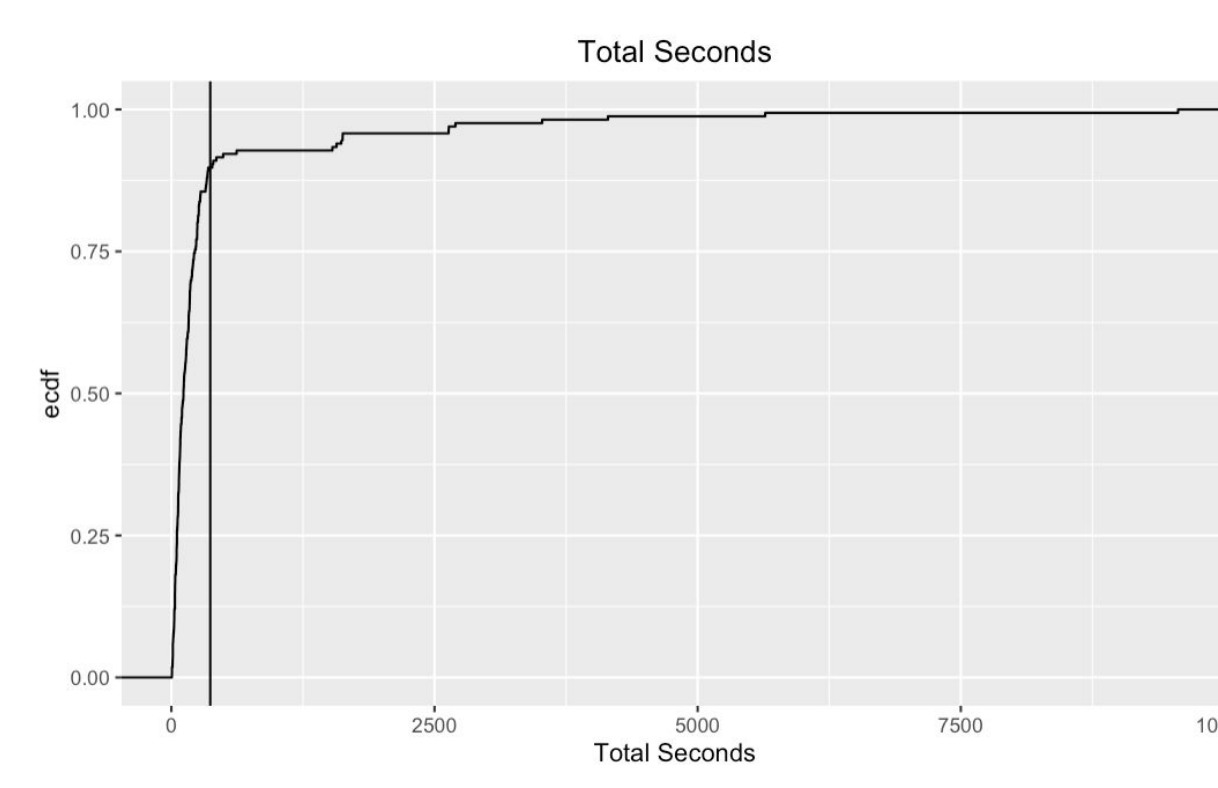


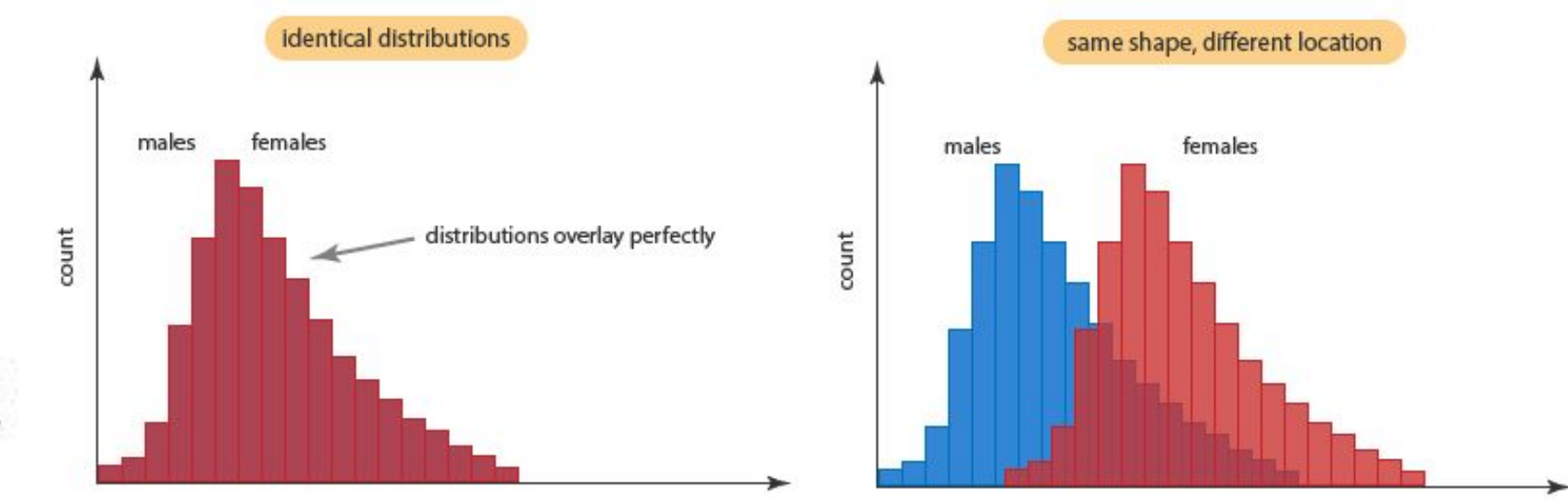
Figure 2: ECDF of Lineup Seconds

METHODS

- We ran seven Mann-Whitney U tests at both the game level and season level to compare lineup types across the different performance metrics
 - Mann-Whitney U test:
$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$
$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

R_1 - sum of ranks for group 1

n_1 - number of observations for group 2
- Due to a small sample size, directionality summaries were combined with test results to describe practical differences.



ANALYSIS & RESULTS

- All p-values were evaluated using an adjusted significance threshold (alpha = 0.01) based on Bonferroni correction.
- After bonferroni correction, none of the tests showed statistically significance evidence that 3G and 4G lineup samples come from populations with different distributions.
- However, individual game-level summaries (example table below) were aggregated across the season to assess directional trends.

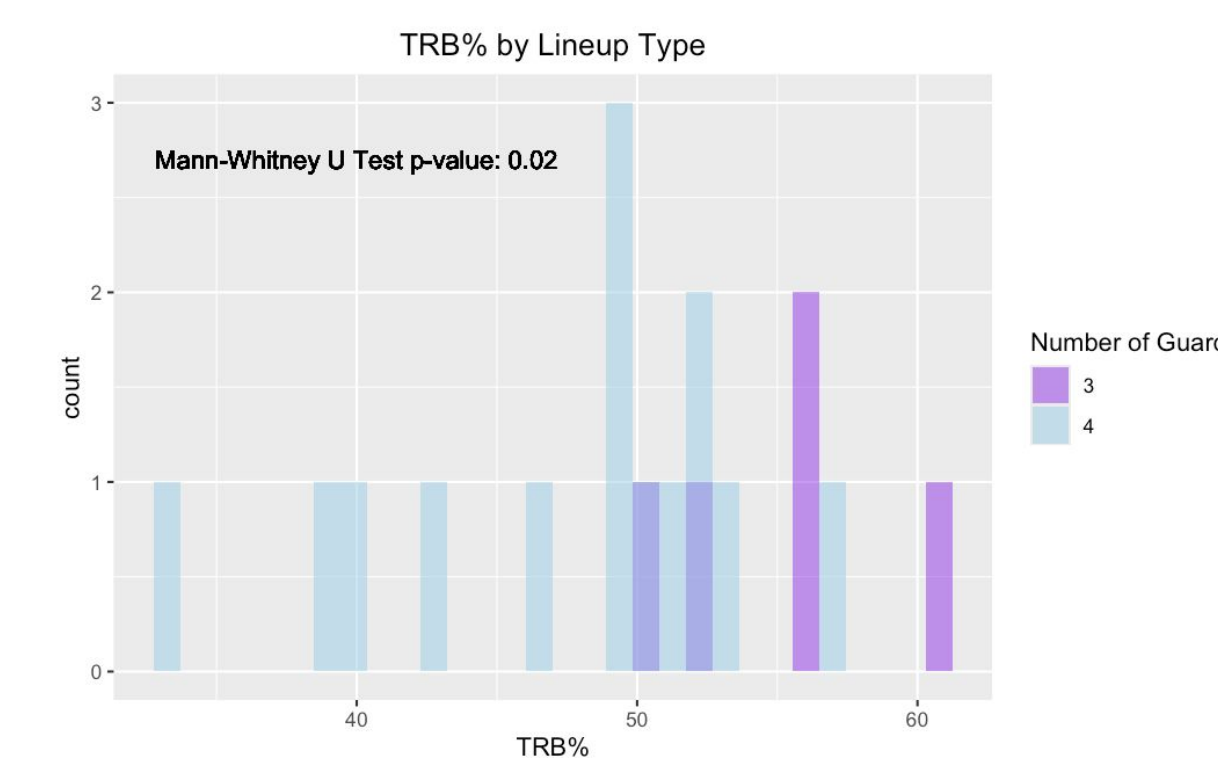


Figure 3: Season level distribution of TRB% by lineup type.

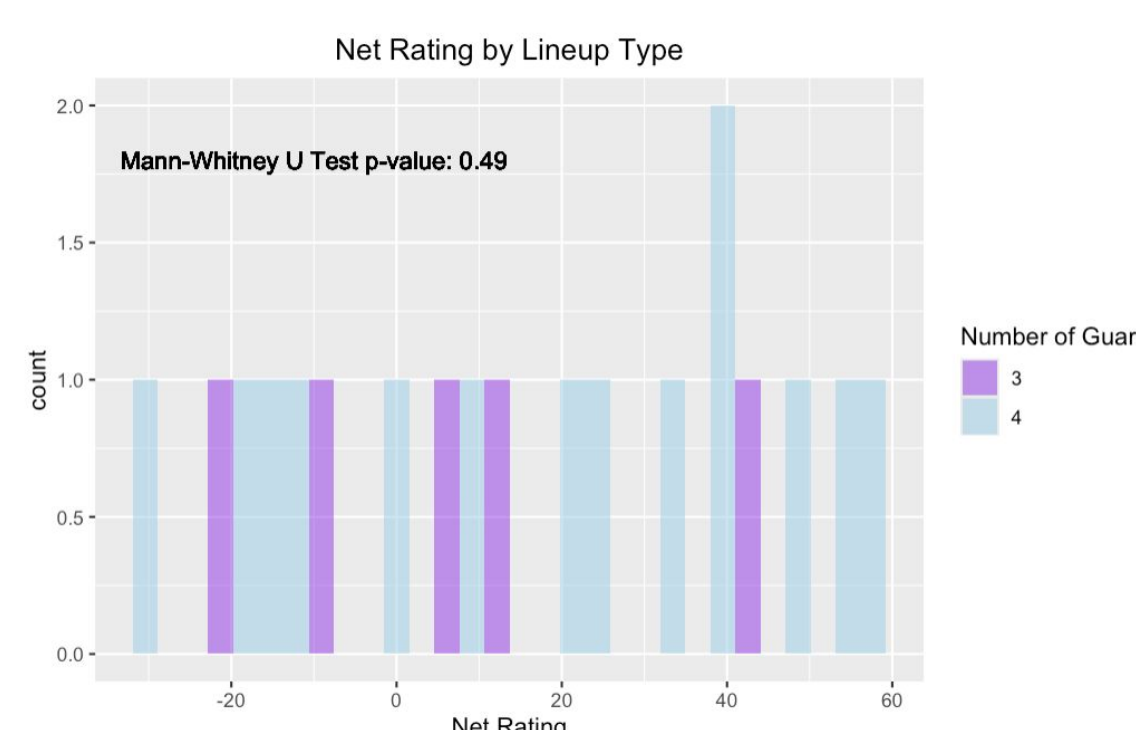


Figure 4: Season level distribution of Net Rating by lineup type.

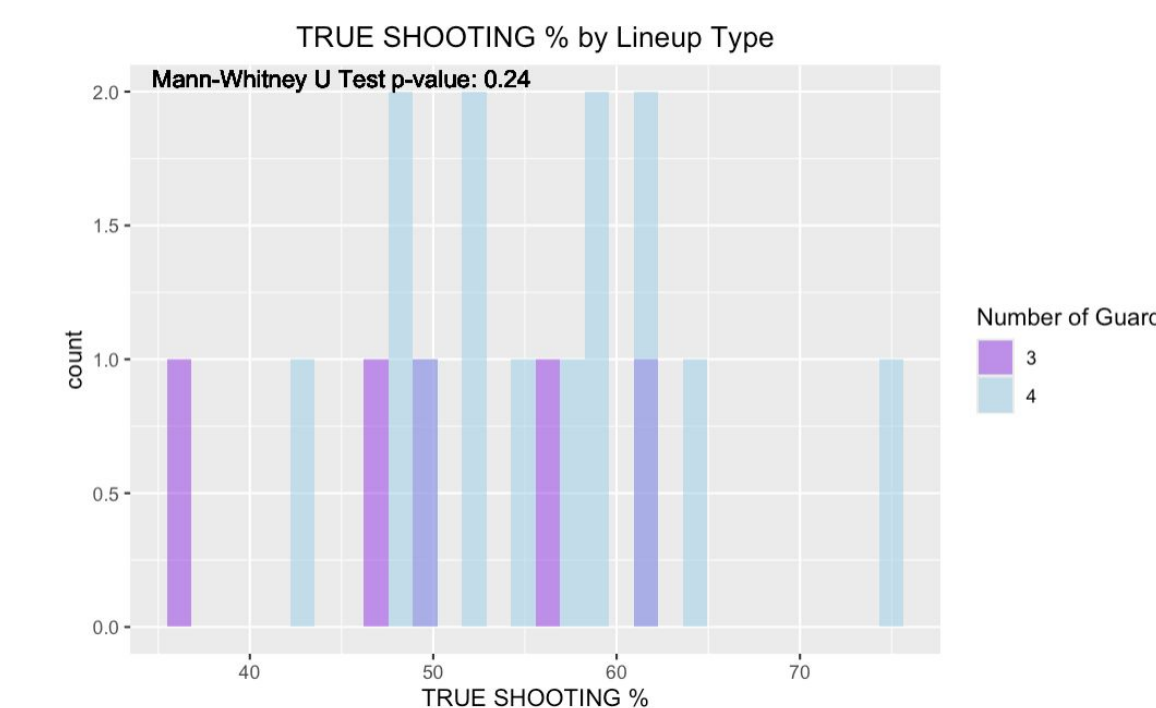


Figure 5: Season level distribution of True Shooting% by lineup type.

Game	Score	W/L	H/A	3G	4G	3G Median Net Rating	4G Median Net Rating	Difference	P-Value
Emory 1	77-67	L	A	4	7	-38.89	0	38.89	0.92
Rochester 1	72-55	W	A	3	4	25	125	100	0.22
Emory 2	78-66	W	H	2	7	-72.71	0	72.71	0.19

Table 1: Snapshot of table of game level directional summaries. Full table includes a row for each game in the 2024-2025 season and columns for the medians, difference, and p-value of each performance metric.

(Blue Highlight - within 15 points, Red highlight - lower value, Red text - loss)

CONCLUSION

- Although none of the seven tests reached statistical significance after Bonferroni correction, consistent directional trends suggest:
 - In over 60% of individual games, 4G lineups showed higher median Net Rating, True Shooting %, and Offensive Rating.
 - In over 50% of individual games, 3G lineups maintained a higher median TRB%.
 - 3PA/FGA and Pace had no decisive factor in lineup success this season.
- Future work could expand the dataset across multiple seasons to assess whether the directionality seen in the 2024-2025 season holds consistently over time

REFERENCES

<https://athletics.cmu.edu/sports/wbkb/2024-25/schedule>

<https://statistics.laerd.com/spss-tutorials/img/mwut/identical-shape-distributions.png>